Notification & Edge-Case QA Report

GetVari hydration app — behavioural verification of all 24 notification triggers and all 17 documented edge cases, and the remediation of every failure it surfaced

REPOSITORY	BRANCH	DATE	RUNNER
getVari / GetVariApp	test-branch (`607c5be`)	31 August 2026	Jest 29 · Node v24.1.0

Command: `npx jest --config jest.config.logic.js`

267	267	0	12
TESTS RUN	PASSED	FAILED	SUITES

1 · What was tested and why

The repository already shipped `__tests__/edgeCases.test.ts`, titled *“Strict Edge-Case Verification (17/17)”* and *“Strict Notification Verification (24/24)”*. It reports 18 green tests, but the coverage is thinner than the title suggests: six of its cases (EC-10, EC-12, EC-13, EC-14, EC-15 and the whole N1–N24 block) have empty bodies containing only a comment, and two more (EC-9, EC-17) assert nothing beyond `toBeDefined()`. Those eight tests cannot go red no matter what the app does.

Three new suites were written to close that gap. They drive the real code paths — a controlled clock, seeded sensor readings, hydration logs, risk scores and persisted cooldown state — and assert that the notification the spec promises is the one that actually reaches `NotificationService.displayNotification`.

SUITE	COVERS	TESTS	PASS	FAIL
`notificationTriggers.test.ts`	N2–N24 firing conditions, cooldowns, quiet hours, permission gate	41	41	0
`edgeCasesStrict.test.ts`	EC-1 – EC-17, executed rather than asserted-by-comment	40	40	0
`reminderScheduling.test.ts`	N1 — the recurring Notifee reminder queue	5	5	0
New coverage added		**86**	**86**	**0**

The remaining 181 tests are the repository’s pre-existing suites. 164 of them passed from the start; 15 were already failing before this work, and two further tests were added while repairing them. All of them pass now — sections 4 and 5 record what was wrong and what changed.

2 · Notification coverage — N1 to N24

All 24 triggers fire under the condition the spec describes. Four delivery defects were found and fixed; the repeat guards and time windows below are the corrected behaviour, and section 4 records what changed.

#	TRIGGER	FIRES WHEN	REPEAT GUARD	TESTS	RESULT
N1	`regular_reminder`	Every 2 min while nothing is logged, queued as Notifee timestamp triggers	Queue reset per log	6	PASS
N2	`low_intake_12pm`	12:00 local and intake below 50% of target	Once per day	2	PASS
N3	`below_target_4pm`	16:00 local and intake below 70% of target	Once per day	1	PASS
N4	`no_water_3h`	Newest log is more than 3 h old	3 h	2	PASS
N5	`goal_achieved`	Daily total reaches the target	Once per day	2	PASS

#	TRIGGER	FIRES WHEN	REPEAT GUARD	TESTS	RESULT
N6	`elevated_risk`	Risk score in 60–79	30 min	1	PASS
N7	`high_risk`	Risk score 80 or above	30 min	1	PASS
N8	`risk_improved`	Score drops below 40 after a risk alert	One per episode	1	PASS
N9	`post_activity`	30+ min at activity load 50 or above	Once per session	2	PASS
N10	`long_activity`	60+ min at activity load 50 or above	Once per session	2	PASS
N11	`high_temp`	Ambient temperature 30–34 °C	3 h	1	PASS
N12	`extreme_heat`	Ambient temperature 35 °C or above	3 h	1	PASS
N13	`high_humidity`	Relative humidity 70% or above	3 h	1	PASS
N14	`inactivity`	3 h at activity load below 10	3 h	1	PASS
N15	`low_battery`	Wearable battery 20% or below	6 h	2	PASS
N16	`fully_charged`	Wearable battery at exactly 100%	Once per day	1	PASS
N17	`morning_msg`	Any time in the 07:00 hour	Once per day	2	PASS
N18	`night_msg`	Any time in the 22:00 hour	Once per day	2	PASS
N19	`weekly_summary`	Monday between 08:00 and 08:59	Once per day	2	PASS
N20	`ai_reminder`	Nothing logged after 10:00	12 h (shared)	1	PASS
N21	`ai_insight`	After 14:00 and below 30% of target	12 h (shared)	1	PASS
N22	`ai_activity`	Activity load above 70	12 h (shared)	1	PASS
N23	`ai_risk`	Risk score above 50	12 h (shared)	1	PASS
N24	`ai_progress`	90% of target reached with risk score below 20	12 h (shared)	1	PASS

Delivery rules

Ten further tests cover the behaviour that sits across all triggers rather than in any one of them. All pass.

RULE	VERIFIED BEHAVIOUR	RESULT
Risk cooldown	A second high-risk alert is suppressed at +10 min and allowed at +35 min	PASS
Environmental cooldown	Extreme heat suppressed at +1 h, allowed at +3 h 30 m	PASS
Battery cooldown	Low battery suppressed at +4 h, allowed at +7 h	PASS
Once per day	Goal achieved is delivered once, then withheld for the rest of the day	PASS
AI spam gate	A different AI nudge 4 h later is held by the shared 12-hour gate	PASS
Quiet hours	At 02:00 the heat, goal and hydration-gap alerts stay silent	PASS
Critical bypass	High risk is delivered at 02:00 with `ignoreSleepWindow: true`	PASS
Permission denied	Nothing is delivered when the OS permission is refused	PASS
Permission denied	Cooldown state still advances, so a later grant does not replay a backlog	PASS
Missing ≠ zero	Null temperature, humidity, battery and activity raise no alert at all	PASS

3 · Edge-case coverage — EC-1 to EC-17

Every one of the seventeen documented edge cases now executes the real code path, and all seventeen behave as specified. EC-14 required implementing the quota switch it documents; see section 4.

#	EDGE CASE	WHAT WAS VERIFIED	TESTS	RESULT
EC–1	Bluetooth disconnected	A score is still produced from 7-day averages; historical step counts convert to an activity load rather than being dropped	2	PASS
EC–2	Sensor failure / impossible reading	HR of 300 and of 12 both report *Device Error*; the shared validator rejects NaN, Infinity and undefined while a real zero survives	3	PASS
EC–3	Stale sensor data	A 20-minute-old reading is stale, a 5-minute-old one is fresh	2	PASS
EC–4	Future-dated intake	An intake an hour ahead is rejected and no row is written	1	PASS
EC–5	Backdated intake	Yesterday's entry is stored with the reported time preserved, distinct from the write time	1	PASS
EC–6	Unrealistic intake	Above 2× the rolling average it needs confirmation; the 1500 mL floor holds; confirming writes it through; zero and negative are rejected	4	PASS
EC–7	Verification tagging	User-reported, sensor-verified and untagged entries all persist the correct provenance	3	PASS
EC–8	Offline / replayed writes	A repeated request id yields one row; concurrent writes share a round trip; a server PK collision reads as duplicate, not error	4	PASS
EC–9	Notifications disabled	Nothing is delivered, but cooldown state still advances	1	PASS
EC–10	Critical alert while asleep	High risk is delivered at 02:00 with the quiet window bypassed, while the routine heat alert stays silent	1	PASS
EC–11	Fluctuating scores	Alert at 65, silence through the 40–60 dead band, one recovery below 40, no second recovery	1	PASS
EC–12	Phone formatted	A wiped device rebuilds the profile from the stored row; a user with no row resolves to null rather than a fabricated profile	2	PASS
EC–13	App reinstalled	Today's 750 mL and the 2800 mL target return from the backend with empty local storage; a read failure degrades to an empty day	2	PASS
EC–14	AI credits exhausted	An exhausted quota degrades each nudge to its rule-based equivalent, restores the AI phrasing when credit returns, and still consumes the 12-hour gate	5	PASS
EC–15	AI spam protection	A second nudge at +6 h is suppressed; one at +13 h is allowed	1	PASS
EC–16	Unknown device state	All-null and two-missing inputs both withhold a score rather than returning zero; a null battery raises neither battery alert	3	PASS
EC–17	Physical damage	Three consecutive dead snapshots read as damage; one good reading clears it; fewer than three, or none, is not enough evidence	4	PASS

4 · Defects found in scope, and the fixes

All five in-scope failures were real gaps in shipped behaviour rather than test noise. Each is now fixed, with the test that proves it.

HIGH · FIXED

F1 — `post_activity` and `long_activity` repeated every 2 seconds

Symptom Once the activity window opened, the notification was re-sent on every evaluation rather than once per session — roughly 900 identical notifications across a 30-minute workout.

Cause `getCooldownFor` had no `case` for either trigger, so both fell through to `default: return 0` and `canSend` reduced to `now - lastSent >= 0` — always true. `HomeScreen.tsx:368` re-runs `evaluate()` on every `telemetry.sensorData` change, and `useHydrationTelemetry.ts:30` sets `TICK_MS = 2000`.

Fix A time-based cooldown is the wrong instrument here — the condition stays true for the whole session. The engine now records `activitySessionAlerts` in its persisted state and clears the list whenever a session starts or ends, so each alert fires exactly once per activity session. The stored blob is spread over defaults on load, so state written by an older build still parses.

Where `src/services/NotificationEngine.ts` — `EnginePersistentState`, the activity-tracking block, the alert block, and `getPersistentState`

MEDIUM · FIXED

F2 — `morning_msg` and `night_msg` only fired in a one-minute window

Symptom A user who opened the app at 07:05 never received the morning greeting, and one who opened it at 22:07 never received the bedtime message, even though neither had been sent that day.

Cause The engine formatted the clock as `"HH:MM"` and compared it for exact equality against `'07:00'` and `'22:00'`. Since the engine only runs while `HomeScreen` is mounted, the app had to be open during that precise minute.

Fix Both now match on the hour, leaning on the `wasSentToday` guard that was already there to prevent repeats — the same arrangement `weekly_summary` uses.

Residual These two still require the app to be open at some point during their hour. Making them independent of the foreground would mean scheduling them as Notifee timestamp triggers, as N1 already is. That is a larger change and was left out of scope.

MEDIUM · FIXED

F3 — EC-14: the AI quota-exhaustion path was unreachable

Symptom The documented behaviour — switch to Rule-Based Mode when the provider returns 429 — could not be triggered or tested.

Cause `triggerAI` contained `const aiQuotaExhausted = false;` annotated “Mock state”, with no setter and no provider error handling, so the branch beneath it was dead code.

Fix The flag is now real engine state with `setAiQuotaExhausted()` / `isAiQuotaExhausted()`, to be driven by whatever sees the 429. When it is set, each nudge degrades to a plain-language equivalent rather than going silent: reminder → *Hydration*, insight → *Below Target*, activity → *Post-Activity*, risk → *Elevated Risk*. `progress` is congratulatory rather than actionable, so with no AI text to send it is dropped. A substitute still consumes the 12-hour spam gate.

Tests The original assertion only checked that a setter existed — the same non-assertion this report criticises elsewhere. It was replaced with four behavioural tests covering the degrade, the per-type mapping, the restore, and the spam gate.

5 · Pre-existing failures, and what they turned out to be

The 15 failures outside the notification and edge-case scope looked like test-setup problems. Repairing the setup revealed that both suites were describing behaviour the shipped code no longer had — the test config was only the outer layer.

11 TESTS · FIXED

F4 — `authSession.test.ts`: the identity scheme had been dropped

First layer The suite could not load: `@react-native-firebase/auth` ships untranspiled ESM and the logic config sets no `transformIgnorePatterns`. Stubbing the two Firebase modules in the suite — the same way every other logic suite mocks its native dependencies — got it running without transpiling a large dependency tree.

Real cause Underneath, an API mismatch. The suite was written against the `AuthService` at commit `8f507b6`, which used `DEMO_PASSCODE = '884200'`, derived a **stable UUID per phone number**, and returned the session from `verifyOtp`. Commit `a1f3458` replaced it with Firebase phone auth, where the demo id is `demo_919876543210` and `verifyOtp` returns `void`. The tests were never updated.

Why the tests were right `getvari_profiles.id` is declared `uuid` in `20260814_create_admin_alerts.sql`, so a `demo_<digits>` id is a write that silently fails — exactly what the suite's comments warn about. `ProfileSheet.tsx` also already read `session.phoneNumber` and `session.isDemo`, the old field names, so the UI was broken against the newer shape too.

Fix `AuthService` was rewritten to merge both: Firebase phone auth stays the primary transport, and the demo-UUID scheme returns as the fallback. The session moved back to AsyncStorage so it survives the cold start a notification tap causes, with a migration that reads any legacy keychain blob once. New Firebase users get a minted UUID profile id rather than their uid, matched on uid then phone number so earlier rows are kept.

4 TESTS · FIXED

F5 — `pbwrMock.test.ts`: wrong backing store, then a real product question

First layer The suite seeded "real" history through its `AsyncStorage` mock, but `TelemetryHistoryService.getHistory` reads from Supabase — AsyncStorage is only the legacy local cache that `syncLocalToCloud` migrates. Every day therefore fell through to `MockTelemetryService`. Adding a `SupabaseClient` mock and seeding through it fixed four of the five.

Real cause The last failure was a genuine conflict. `PBWRService` computed `isCalibrationPeriod` as `snapshots.every(isMock)` and zeroed the activity, temperature and humidity adjustments — so any user whose wearable had not yet reported real telemetry was pinned to zero adjustments indefinitely, which is the opposite of what the mock fallback exists for. The code carried an unfinished note to that effect.

Fix Calibration is now measured from the account's age, as the code's own `created_at` comment intended: a new `accountCreatedAt` field is stamped on the first profile save, and adjustments are withheld for seven days from it. Rows predating the stamp are treated as established, since they are long past day seven. Two tests were added for the inside- and outside-the-window cases.

6 · What still needs a decision

The suite is green, but the following were deliberately left for you rather than settled here.

#	OPEN ITEM	PRIORITY
1	**Demo id migration.** Restoring the UUID scheme means any rows already written under a `demo_<digits>` id by the Firebase-only build are orphaned. If that build reached real devices, those rows need remapping before release.	High

#	OPEN ITEM	PRIORITY
2	**Schema disagreement.** `20260814_create_admin_alerts.sql` declares `getvari_profiles.id` as `uuid`; `admin_setup.sql` declares the referencing column as `text` “to match”. One of the two is wrong and should be reconciled.	High
3	**Foreground dependency.** The greetings, the weekly summary and every rule-based trigger only run while the dashboard is mounted. Only N1 survives the app being closed. Scheduling the time-based ones as Notifee triggers would be the durable fix.	Medium
4	**Redundant suite.** `edgeCases.test.ts` still contains the eight non-asserting tests. Now that `edgeCasesStrict.test.ts` covers the same ground for real, the older file can be deleted so the coverage numbers mean something.	Low

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Generated from a full run of `jest.config.logic.js` on branch `test-branch`, 31 August 2026. 267 tests across 12 suites · 267 passed · 0 failed. New suites: `__tests__/notificationTriggers.test.ts`, `__tests__/edgeCasesStrict.test.ts`, `__tests__/reminderScheduling.test.ts`. Services changed: `AuthService`, `NotificationEngine`, `PBWRService`, `ProfileService`.